

NexusAI — Enterprise AI Operations Platform

Database Schema Specification · v1.0

PostgreSQL 16 · pgvector · SQLAlchemy 2.0 + Alembic 55 tables · 8 domains · Multi-tenant · Append-only audit · Effective-dated pricing

Design principles & global conventions

- **UUID primary keys everywhere** — `gen_random_uuid()` via pgcrypto. Prevents enumeration and lets a client generate an ID before insert. The two exceptions are `audit_log` and `spans`, which use `BIGSERIAL` because insertion order carries meaning and volume is high.
- **All timestamps `TIMESTAMPZ`** — stored in UTC without exception. Local time conversion happens in the UI layer only, never in a query.
- **Workspace scoping on every business table** — `workspace_id` is present and indexed on every table a user can reach, including denormalised onto hot child tables. Multi-tenancy enforced by a mandatory predicate in the repository layer; a query without it is a bug caught by test, not by review.
- **Immutable tables** — `audit_log`, `cost_entries`, `policy_violations`, `agent_steps`, `eval_results`, `pii_detections`, `fallback_events` are append-only. Enforced by Postgres row-level security, not application convention.
- **Soft deletes on configuration, hard deletes on nothing user-facing** — `deleted_at` on definitional tables (agents, prompts, collections, policies). Historical runs must still resolve their configuration after it is retired.
- **`VARCHAR` + `CHECK`, never native `ENUM`** — `ALTER TYPE` takes a table lock; adding a value to a `CHECK` constraint does not. SQLAlchemy generates Python enums from the same constants.
- **`NUMERIC` for money, never float** — `NUMERIC(14,8)` for per-request cost, `NUMERIC(14,4)` for aggregates, `NUMERIC(12,6)` for per-million-token rates. Individual request costs are fractions of a cent and float error accumulates visibly across a month.
- **Effective-dated pricing** — `pricing_rates` is versioned by `effective_from`. Cost is computed against the rate live at request time, so historical reports never change when a provider updates prices.
- **JSONB for open shapes, columns for anything queried** — configuration, tool arguments and decision factors are JSONB. Anything filtered, grouped or aggregated gets a real column.
- **`updated_at` by trigger** — one shared `set_updated_at()` function applied to every table with the column. Correct even for direct SQL and migrations.
- **Vectors are versioned** — `chunk_embeddings` carries the model and embedding version on every row, which is what makes an embedding-model migration possible without downtime.
- **Partitioning where volume demands it** — `audit_log` and `traces` by month, `spans` and `gateway_requests` by day. Partition creation is a scheduled job, not a manual chore.

Column flag legend			
Flag	Meaning	Flag	Meaning
PK	Primary key	FK	Foreign key, referenced table in description
NN	NOT NULL	UQ	UNIQUE constraint
IDX	Indexed, non-unique	DEF	Has a default, stated in description

Schema summary — 55 tables across 8 domains

Domain	#	Tables
1. Identity & Access	7	users, workspaces, workspace_memberships, roles, role_assignments, api_keys, user_sessions
2. Gateway & Routing	8	providers, models, model_deployments, routing_policies, routing_rules, gateway_requests, gateway_messages, fallback_events
3. Cost & Quotas	5	pricing_rates, cost_entries, budgets, quota_windows, rate_limit_buckets
4. Knowledge Hub	9	collections, collection_grants, documents, document_versions, chunks, chunk_embeddings, ingestion_jobs, retrieval_queries, retrieval_citations
5. Agents & Workflows	7	agent_definitions, agent_versions, tools, agent_tool_grants, agent_runs, agent_steps, approval_requests
6. Evaluation	7	prompt_templates, prompt_versions, eval_datasets, eval_items, scorers, eval_runs, eval_results
7. Observability	5	traces, spans, span_events, metric_rollups, alert_rules
8. Governance & Audit	7	audit_log, policies, policy_violations, pii_detections, redaction_rules, feature_flags, data_retention_schedules

1. Identity & Access

users

Platform identities, provisioned on first SSO login.

Column	Type	Flags	Description
id	UUID	PK	gen_random_uuid()
email	VARCHAR(320)	UQ NN IDX	Lowercased and trimmed on insert
external_subject	VARCHAR(255)	UQ IDX	OIDC sub claim. Stable across email changes — match on this, not email
issuer	VARCHAR(255)	NN	OIDC issuer URL. Same subject from a different issuer is a different person
display_name	VARCHAR(200)	NN	
avatar_url	VARCHAR(500)		
default_workspace_id	UUID	FK	→ workspaces.id. Landing workspace after login
clearance	VARCHAR(20)	NN	public internal confidential restricted. Matched against document sensitivity in retrieval. DEF internal
is_active	BOOLEAN	NN	Deactivated users cannot authenticate. Never hard-delete. DEF true
last_login_at	TIMESTAMP TZ		
created_at	TIMESTAMP TZ	NN	
updated_at	TIMESTAMP TZ	NN	Trigger-maintained

Indexes `uq_users_email (email)` · `uq_users_subject (issuer, external_subject)` · `idx_users_active (is_active) PARTIAL WHERE is_active`

workspaces

The tenancy boundary. Every business record belongs to exactly one.

Column	Type	Flags	Description
id	UUID	PK	
slug	VARCHAR(60)	UQ NN	URL-safe identifier
name	VARCHAR(200)	NN	
description	TEXT		
logging_mode	VARCHAR(20)	NN	full redacted metadata_only. Governs what gateway_messages stores. DEF redacted
default_routing_policy_id	UUID	FK	→ routing_policies.id. Used when a request specifies neither model nor policy
monthly_budget_usd	NUMERIC(12,2)		NULL = unlimited. Enforced by GW-007
data_residency	VARCHAR(20)		any eu us local_only. Constrains which provider deployments are eligible
allow_external_providers	BOOLEAN	NN	FALSE forces routing to self-hosted deployments only. DEF true
settings	JSONB		Open configuration bag for module-level defaults
is_active	BOOLEAN	NN	DEF true
created_at	TIMESTAMP TZ	NN	
updated_at	TIMESTAMP TZ	NN	

workspace_memberships

Which users belong to which workspaces.

Column	Type	Flags	Description
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id	UUID	PK	
workspace_id	UUID	FK NN IDX	→ workspaces.id ON DELETE CASCADE
user_id	UUID	FK NN IDX	→ users.id ON DELETE CASCADE
invited_by	UUID	FK	→ users.id
joined_at	TIMESTAMPTZ	NN	
created_at	TIMESTAMPTZ	NN	

Indexes `uq_membership (workspace_id, user_id) UNIQUE`

roles

Named permission bundles. Permissions live in JSONB so a role extends without a migration.

Column	Type	Flags	Description
id	UUID	PK	
workspace_id	UUID	FK IDX	→ workspaces.id. NULL = system role available to all workspaces
name	VARCHAR(50)	NN	owner admin engineer analyst viewer, or custom
description	TEXT		
permissions	JSONB	NN	Array of permission strings, e.g. ["gateway.route.write", "knowledge.collection.read"]
is_system	BOOLEAN	NN	System roles cannot be edited or deleted. DEF false
created_at	TIMESTAMPTZ	NN	
updated_at	TIMESTAMPTZ	NN	

Indexes `uq_role_name (workspace_id, name) UNIQUE NULLS NOT DISTINCT`

role_assignments

Binds a user to a role within a workspace.

Column	Type	Flags	Description
id	UUID	PK	
workspace_id	UUID	FK NN IDX	→ workspaces.id
user_id	UUID	FK NN IDX	→ users.id
role_id	UUID	FK NN	→ roles.id
granted_by	UUID	FK NN	→ users.id. Audited
expires_at	TIMESTAMPTZ		Time-bounded elevation. NULL = permanent
created_at	TIMESTAMPTZ	NN	

Indexes `uq_role_assignment (workspace_id, user_id, role_id) UNIQUE · idx_role_expiry (expires_at) PARTIAL WHERE expires_at IS NOT NULL`

api_keys

Service credentials. Scoped to one workspace, hashed at rest, shown once.

Column	Type	Flags	Description
id	UUID	PK	
workspace_id	UUID	FK NN IDX	→ workspaces.id ON DELETE CASCADE
key_prefix	VARCHAR(16)	NN	First 12 chars for display, e.g. nx_live_A7f2. Identifies a key without exposing it
key_hash	VARCHAR(64)	UQ NN	SHA-256 of the full key. Plaintext never stored
environment	VARCHAR(10)	NN	live test ✓ CHECK
label	VARCHAR(100)	NN	Human name, e.g. support-bot-prod
scopes	TEXT[]		Permission scopes. NULL = the workspace's default service scope

created_by	UUID	FK NN	→ users.id
last_used_at	TIMESTAMPTZ		Written at most once per minute to limit write amplification
last_used_ip	INET		Anomaly detection
request_count	BIGINT	NN	DEF 0
expires_at	TIMESTAMPTZ		NULL = never. Set to force rotation
revoked_at	TIMESTAMPTZ	IDX	NULL = active
revoked_by	UUID	FK	→ users.id
created_at	TIMESTAMPTZ	NN	

Indexes `idx_apikey_hash (key_hash) UNIQUE` · `idx_apikey_ws (workspace_id, revoked_at)`

user_sessions

Browser sessions with revocation state.

Column	Type	Flags	Description
id	UUID	PK	
user_id	UUID	FK NN IDX	→ users.id
token_hash	VARCHAR(64)	UQ NN	SHA-256 of the opaque session token
workspace_id	UUID	FK	Active workspace at last request
ip_address	INET		
user_agent	TEXT		
status	VARCHAR(20)	NN IDX	active expired revoked logged_out
revoke_reason	VARCHAR(50)		user_logout admin_force suspicious_activity password_change
expires_at	TIMESTAMPTZ	NN IDX	8h default, sliding refresh capped at 24h
last_active_at	TIMESTAMPTZ		
created_at	TIMESTAMPTZ	NN	

Indexes `idx_session_token (token_hash) UNIQUE` · `idx_session_expiry (expires_at)` — cleanup job

2. Gateway & Routing

providers

Upstream model providers. Adding one is configuration, not code.

Column	Type	Flags	Description
id	UUID	PK	
slug	VARCHAR(50)	UQ NN	openai anthropic azure_openai vertex local mock
name	VARCHAR(100)	NN	
adapter_class	VARCHAR(200)	NN	Import path of the <code>LLMProvider</code> implementation
base_url	VARCHAR(500)		Override for self-hosted or regional endpoints
credential_ref	VARCHAR(200)		Key into the secret resolver. Never the credential itself
is_external	BOOLEAN	NN	FALSE for self-hosted. Data policies gate on this. DEF true
region	VARCHAR(20)		Residency enforcement
default_timeout_ms	INTEGER	NN	DEF 60000
max_concurrency	SMALLINT	NN	Connection pool ceiling. DEF 20
health_status	VARCHAR(20)	NN	healthy degraded unavailable unknown. Written by the health probe
health_checked_at	TIMESTAMPTZ		
is_active	BOOLEAN	NN	DEF true
created_at	TIMESTAMPTZ	NN	
updated_at	TIMESTAMPTZ	NN	

models

A model as the platform knows it, independent of where it is deployed.

Column	Type	Flags	Description
id	UUID	PK	
slug	VARCHAR(100)	UQ NN IDX	Platform-wide name used in API requests, e.g. <code>fast-general</code> , <code>strong-reasoning</code>
provider_id	UUID	FK NN IDX	→ providers.id
provider_model_name	VARCHAR(200)	NN	The name the provider expects on the wire
family	VARCHAR(50)		Groups variants for comparison in the Evaluation Lab
tier	VARCHAR(20)	NN IDX	cheap standard strong embedding. Routing rules reference this
modality	VARCHAR(20)	NN	text multimodal embedding
context_window	INTEGER	NN	Max input tokens. Used for context-length fallback decisions
max_output_tokens	INTEGER		
supports_tools	BOOLEAN	NN	DEF false
supports_streaming	BOOLEAN	NN	DEF true
supports_json_mode	BOOLEAN	NN	DEF false
embedding_dimensions	SMALLINT		Only for embedding models. Must match the vector column width
is_active	BOOLEAN	NN	Inactive models cannot be selected by new requests but remain resolvable for history
deprecated_at	TIMESTAMPTZ		Warning surfaced on routing policies still referencing it
created_at	TIMESTAMPTZ	NN	

updated_at	TIMESTAMPTZ	NN
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Indexes `idx_models_slug (slug) UNIQUE · idx_models_tier (tier, is_active)`

model_deployments

The same model in different regions or on different endpoints. Residency policies select here.

Column	Type	Flags	Description
id	UUID	PK	
model_id	UUID	FK NN IDX	→ models.id
region	VARCHAR(20)	NN	eu us local
endpoint_url	VARCHAR(500)		Overrides the provider base URL
deployment_name	VARCHAR(200)		Azure deployment name or equivalent
priority	SMALLINT	NN	Lower is preferred among eligible deployments. DEF 100
is_active	BOOLEAN	NN	DEF true
created_at	TIMESTAMPTZ	NN	

Indexes `idx_deploy_model_region (model_id, region, is_active)`

routing_policies

A named, versioned routing configuration.

Column	Type	Flags	Description
id	UUID	PK	
workspace_id	UUID	FK NN IDX	→ workspaces.id
name	VARCHAR(100)	NN	
version	INTEGER	NN	Incremented on change; prior versions retained
is_current	BOOLEAN	NN	One current version per name per workspace
default_model_id	UUID	FK NN	→ models.id. Used when no rule matches
fallback_model_ids	UUID[]		Ordered preference list for GW-004
max_cost_per_request_usd	NUMERIC(10,6)		Hard ceiling. Requests estimated above it are rejected
total_timeout_ms	INTEGER	NN	Wall-clock budget across all retries and fallbacks. DEF 60000
description	TEXT		
created_by	UUID	FK NN	→ users.id
deleted_at	TIMESTAMPTZ		
created_at	TIMESTAMPTZ	NN	

Indexes `uq_policy_current (workspace_id, name, version) UNIQUE · idx_policy_current (workspace_id, name, is_current) PARTIAL WHERE is_current`

routing_rules

Ordered conditions within a policy. First match wins.

Column	Type	Flags	Description
id	UUID	PK	
routing_policy_id	UUID	FK NN IDX	→ routing_policies.id ON DELETE CASCADE
sequence	SMALLINT	NN	Evaluation order, 1-indexed
name	VARCHAR(100)	NN	Shown in the request record so a routing decision is self-explaining
condition	JSONB	NN	e.g. {"all":

```
[[{"field": "metadata.task_type", "op": "in", "value": ["summarise"]}, {"field": "estimated_input_tokens", "op": "lt", "value": 4000}]]
```

action	VARCHAR(20)	NN	route route_preference_list reject ✓ CHECK
target_model_id	UUID	FK	→ models.id. Required when action = route
target_model_ids	UUID[]		Ordered list when action = route_preference_list
reject_reason	VARCHAR(200)		Returned to the caller when action = reject
is_active	BOOLEAN	NN	DEF true
created_at	TIMESTAMPTZ	NN	

Indexes `uq_rule_seq` (routing_policy_id, sequence) **UNIQUE**

gateway_requests

Every request through the gateway. Partitioned by day.

Column	Type	Flags	Description
id	UUID	PK	Also the public <code>x-nexus-request-id</code>
workspace_id	UUID	FK NN IDX	→ workspaces.id
user_id	UUID	FK IDX	→ users.id. NULL for service-key requests
api_key_id	UUID	FK IDX	→ api_keys.id. NULL for user requests
agent_run_id	UUID	FK IDX	→ agent_runs.id when the caller was an agent
eval_run_id	UUID	FK IDX	→ eval_runs.id when the caller was an evaluation
routing_policy_id	UUID	FK	→ routing_policies.id. NULL when the model was pinned
routing_rule_id	UUID	FK	→ routing_rules.id. Which rule decided, for explainability
requested_model_id	UUID	FK	→ models.id. What the caller asked for, if anything
resolved_model_id	UUID	FK IDX	→ models.id. What actually served the request
prompt_version_id	UUID	FK	→ prompt_versions.id when a registered prompt was used
status	VARCHAR(20)	NN IDX	<code>in_flight</code> <code>success</code> <code>error</code> <code>rejected</code> <code>cancelled</code>
error_class	VARCHAR(40)	IDX	Normalised: <code>rate_limit</code> <code>timeout</code> <code>context_length</code> <code>content_filter</code> <code>auth</code> <code>unavailable</code> <code>policy_denied</code> <code>quota_exceeded</code>
error_detail	TEXT		
input_tokens	INTEGER		
output_tokens	INTEGER		
cached_tokens	INTEGER		Prompt-cache hits, priced differently
estimated_input_tokens	INTEGER		Pre-call estimate used for routing. Kept to measure estimator drift
fallback_count	SMALLINT	NN	DEF 0
retry_count	SMALLINT	NN	DEF 0
duration_ms	INTEGER		Total, including retries and fallbacks
provider_duration_ms	INTEGER		Provider time only. The difference is gateway overhead
trace_id	VARCHAR(32)	IDX	OTel trace ID — joins to <code>traces</code>
idempotency_key	VARCHAR(255)	IDX	Replay key, 24h validity
metadata	JSONB		Caller-supplied tags used by routing rules and reporting
is_streaming	BOOLEAN	NN	DEF false
created_at	TIMESTAMPTZ	NN IDX	Partition key, by day
completed_at	TIMESTAMPTZ		

Indexes `idx_gwreq_ws_time` (`workspace_id`, `created_at` DESC) · `idx_gwreq_model` (`resolved_model_id`, `created_at` DESC) · `idx_gwreq_status` (`status`, `error_class`) PARTIAL WHERE `status <> 'success'` · `idx_gwreq_idem` (`idempotency_key`, `workspace_id`) UNIQUE PARTIAL WHERE `idempotency_key IS NOT NULL`

gateway_messages

Prompt and completion content, subject to the workspace logging mode.

Column	Type	Flags	Description
id	UUID	PK	
gateway_request_id	UUID	FK NN IDX	→ gateway_requests.id ON DELETE CASCADE
workspace_id	UUID	FK NN IDX	Denormalised for retention sweeps
sequence	SMALLINT	NN	Message order within the request
role	VARCHAR(20)	NN	system user assistant tool ✓ CHECK
content	TEXT		NULL in <code>metadata_only</code> mode, redacted in <code>redacted</code> mode
content_ref	VARCHAR(500)		Object storage key when content exceeds 8KB
content_hash	VARCHAR(64)		SHA-256 of the original, retained even when content is not
token_count	INTEGER		
tool_calls	JSONB		Structured tool calls emitted by the assistant
redaction_applied	BOOLEAN	NN	DEF false
purge_after	DATE	IDX	Set from <code>data_retention_schedules</code> . Content nulled after this date
created_at	TIMESTAMPTZ	NN	

Indexes `uq_msg_seq` (`gateway_request_id`, `sequence`) UNIQUE · `idx_msg_purge` (`purge_after`) PARTIAL WHERE `content IS NOT NULL`

fallback_events

Every time routing moved a request to a different model. Append-only.

Column	Type	Flags	Description
id	UUID	PK	
gateway_request_id	UUID	FK NN IDX	→ gateway_requests.id
workspace_id	UUID	FK NN IDX	Denormalised
attempt_number	SMALLINT	NN	1-indexed
from_model_id	UUID	FK NN	→ models.id
to_model_id	UUID	FK	→ models.id. NULL when the fallback chain was exhausted
trigger_error_class	VARCHAR(40)	NN IDX	Normalised error that caused it
provider_status_code	SMALLINT		
circuit_breaker_open	BOOLEAN	NN	TRUE when the source model was skipped by an open breaker rather than tried. DEF false
latency_before_fallback_ms	INTEGER		Time wasted on the failed attempt
created_at	TIMESTAMPTZ	NN IDX	IMMUTABLE

Indexes `idx_fallback_models` (`from_model_id`, `to_model_id`, `created_at` DESC)

3. Cost & Quotas

pricing_rates

Effective-dated model pricing. The reason historical cost reports stay correct.

Column	Type	Flags	Description
id	UUID	PK	
model_id	UUID	FK NN IDX	→ models.id
input_per_million_usd	NUMERIC(12,6)	NN	
output_per_million_usd	NUMERIC(12,6)	NN	
cached_input_per_million_usd	NUMERIC(12,6)		Prompt-cache rate. NULL = same as input
currency	CHAR(3)	NN	DEF USD
effective_from	TIMESTAMP TZ	NN IDX	
effective_to	TIMESTAMP TZ		NULL = currently in force
source	VARCHAR(50)	NN	provider_published negotiated estimated
notes	TEXT		
created_by	UUID	FK	→ users.id
created_at	TIMESTAMP TZ	NN	

Indexes `idx_rate_lookup (model_id, effective_from DESC)` · EXCLUDE constraint preventing overlapping `[effective_from, effective_to)` ranges per model — overlapping rates make cost ambiguous and the database should refuse them.

cost_entries

One row per billable operation. Append-only.

Column	Type	Flags	Description
id	UUID	PK	
workspace_id	UUID	FK NN IDX	→ workspaces.id
gateway_request_id	UUID	FK IDX	→ gateway_requests.id
user_id	UUID	FK IDX	→ users.id
api_key_id	UUID	FK IDX	→ api_keys.id
agent_run_id	UUID	FK IDX	→ agent_runs.id
eval_run_id	UUID	FK IDX	→ eval_runs.id
ingestion_job_id	UUID	FK IDX	→ ingestion_jobs.id. Populated for embedding cost
model_id	UUID	FK NN IDX	→ models.id
pricing_rate_id	UUID	FK NN	→ pricing_rates.id. The exact rate applied
operation	VARCHAR(20)	NN IDX	completion embedding rerank judge. Keeps ingestion cost separable from query cost
input_tokens	INTEGER	NN	DEF 0
output_tokens	INTEGER	NN	DEF 0
cached_tokens	INTEGER	NN	DEF 0
cost_usd	NUMERIC(14,8)	NN	Computed at write time, never recomputed
accounting_period	CHAR(7)	NN IDX	YYYY-MM. Monthly aggregation key
created_at	TIMESTAMP TZ	NN IDX	IMMUTABLE

Indexes `idx_cost_ws_period (workspace_id, accounting_period, operation)` · `idx_cost_model_time (model_id, created_at DESC)`

budgets

Spend ceilings per workspace per period.

Column	Type	Flags	Description
id	UUID	PK	
workspace_id	UUID	FK NN IDX	→ workspaces.id
period_type	VARCHAR(10)	NN	monthly daily ✓ CHECK
amount_usd	NUMERIC(12,2)	NN	
soft_threshold_pct	SMALLINT	NN	Warning point. DEF 80
hard_stop	BOOLEAN	NN	TRUE rejects requests at 100%; FALSE only alerts. DEF true
alert_channel_ids	UUID[]		→ alert_rules.id for notification routing
effective_from	DATE	NN	
effective_to	DATE		
is_active	BOOLEAN	NN	DEF true
created_at	TIMESTAMPTZ	NN	
updated_at	TIMESTAMPTZ	NN	

quota_windows

Usage counters per workspace per window. Survives a Redis flush.

Column	Type	Flags	Description
id	UUID	PK	
workspace_id	UUID	FK NN IDX	→ workspaces.id
window_type	VARCHAR(10)	NN	minute hour day month
window_start	TIMESTAMPTZ	NN IDX	
request_count	INTEGER	NN	DEF 0
input_tokens	BIGINT	NN	DEF 0
output_tokens	BIGINT	NN	DEF 0
cost_usd	NUMERIC(14,6)	NN	DEF 0
rejected_rate_limit	INTEGER	NN	DEF 0
rejected_budget	INTEGER	NN	DEF 0
updated_at	TIMESTAMPTZ	NN	

Indexes `uq_quota_window (workspace_id, window_type, window_start) UNIQUE`

rate_limit_buckets

Configured limits. Live counters are in Redis; this table is the configuration and the console's view of it.

Column	Type	Flags	Description
id	UUID	PK	
workspace_id	UUID	FK NN IDX	→ workspaces.id
api_key_id	UUID	FK IDX	→ api_keys.id. NULL = workspace-wide limit
scope	VARCHAR(20)	NN	workspace api_key user
requests_per_minute	INTEGER		NULL = unlimited
tokens_per_minute	INTEGER		
concurrent_requests	SMALLINT		
is_active	BOOLEAN	NN	DEF true
created_at	TIMESTAMPTZ	NN	
updated_at	TIMESTAMPTZ	NN	

4. Knowledge Hub

collections

A governed set of documents with one embedding configuration.

Column	Type	Flags	Description
id	UUID	PK	
workspace_id	UUID	FK NN IDX	→ workspaces.id
slug	VARCHAR(80)	NN	
name	VARCHAR(200)	NN	
description	TEXT		
embedding_model_id	UUID	FK NN	→ models.id. Must be an embedding model
embedding_version	SMALLINT	NN	Incremented on re-embed. chunk_embeddings filters on it. DEF 1
active_embedding_version	SMALLINT	NN	Flipped atomically after a re-embed completes. Zero-downtime migration hinges on these being separate
chunking_strategy	VARCHAR(20)	NN	recursive semantic structural ✓ CHECK
chunk_size_tokens	SMALLINT	NN	DEF 800
chunk_overlap_tokens	SMALLINT	NN	DEF 120
rerank_enabled	BOOLEAN	NN	Cross-encoder rerank. Costs latency, so it is a decision. DEF false
default_sensitivity	VARCHAR(20)	NN	Applied to documents that do not set their own. DEF internal
freshness_window_days	SMALLINT		NULL = documents never go stale
exclude_stale	BOOLEAN	NN	Remove stale documents from retrieval entirely. DEF false
document_count	INTEGER	NN	Denormalised. DEF 0
chunk_count	INTEGER	NN	Denormalised. DEF 0
last_indexed_at	TIMESTAMPTZ		
deleted_at	TIMESTAMPTZ		
created_at	TIMESTAMPTZ	NN	
updated_at	TIMESTAMPTZ	NN	

Indexes `uq_collection_slug (workspace_id, slug) UNIQUE`

collection_grants

Who may read or manage a collection. Retrieval joins against this table — it is a hot path, not an afterthought.

Column	Type	Flags	Description
id	UUID	PK	
collection_id	UUID	FK NN IDX	→ collections.id ON DELETE CASCADE
workspace_id	UUID	FK NN IDX	Denormalised for the retrieval predicate
grantee_type	VARCHAR(10)	NN	role user ✓ CHECK
role_id	UUID	FK	→ roles.id
user_id	UUID	FK	→ users.id
access_level	VARCHAR(10)	NN	read manage ✓ CHECK
granted_by	UUID	FK NN	→ users.id. Every grant change is audited
expires_at	TIMESTAMPTZ		
created_at	TIMESTAMPTZ	NN	

Indexes `idx_grant_lookup (collection_id, grantee_type, role_id, user_id)` · `idx_grant_user (user_id, collection_id)` — the index the retrieval

documents

A logical document. Content lives in `document_versions`.

Column	Type	Flags	Description
id	UUID	PK	
collection_id	UUID	FK NN IDX	→ collections.id ON DELETE CASCADE
workspace_id	UUID	FK NN IDX	Denormalised
title	VARCHAR(500)	NN	
source_type	VARCHAR(20)	NN	upload url connector
source_uri	VARCHAR(1000)		Original location, for re-fetch and freshness checks
sensitivity	VARCHAR(20)	NN	public internal confidential restricted. Matched against users.clearance
current_version_id	UUID	FK	→ document_versions.id
author	VARCHAR(200)		
document_date	DATE		Business date of the content, distinct from upload date. Freshness uses this
metadata	JSONB		Arbitrary filterable attributes: department, doc_type, product
is_stale	BOOLEAN	NN	Set by the freshness job. DEF false
stale_reason	VARCHAR(50)		source_changed past_freshness_window
deleted_at	TIMESTAMPTZ		
created_at	TIMESTAMPTZ	NN	
updated_at	TIMESTAMPTZ	NN	

Indexes `idx_docs_collection (collection_id, deleted_at)` · `idx_docs_metadata (metadata) USING GIN` · `idx_docs_stale (is_stale) PARTIAL WHERE is_stale`

document_versions

Immutable content revisions. A re-upload versions rather than overwrites.

Column	Type	Flags	Description
id	UUID	PK	
document_id	UUID	FK NN IDX	→ documents.id ON DELETE CASCADE
version_number	SMALLINT	NN	1-indexed
storage_key	VARCHAR(500)	NN	Object storage key. Never a public URL; signed URLs generated on demand
content_hash	VARCHAR(64)	NN IDX	SHA-256. Deduplication and change detection
mime_type	VARCHAR(100)	NN	
size_bytes	BIGINT	NN	
page_count	SMALLINT		
extracted_text_key	VARCHAR(500)		Parsed plain text, cached to avoid reparsing on re-chunk
uploaded_by	UUID	FK NN	→ users.id
is_current	BOOLEAN	NN	DEF true
created_at	TIMESTAMPTZ	NN	IMMUTABLE

Indexes `uq_docver (document_id, version_number) UNIQUE` · `idx_docver_hash (content_hash)`

chunks

Retrievable units with full provenance. Provenance quality is citation quality.

Column	Type	Flags	Description
id	UUID	PK	
document_version_id	UUID	FK NN IDX	→ document_versions.id ON DELETE CASCADE
document_id	UUID	FK NN IDX	Denormalised
collection_id	UUID	FK NN IDX	Denormalised — retrieval filters on it before any join
workspace_id	UUID	FK NN IDX	Denormalised
sequence	INTEGER	NN	Order within the document
content	TEXT	NN	Raw chunk text, displayed in citations
embedded_text	TEXT	NN	Content prefixed with title and heading path. This is what was embedded, and it differs from <code>content</code> deliberately
token_count	SMALLINT	NN	
page_number	SMALLINT		
section_path	TEXT[]		Heading hierarchy, e.g. {Policies, Leave, Approval}
char_start	INTEGER		Offset in the source text, for highlighting
char_end	INTEGER		
chunking_strategy	VARCHAR(20)	NN	The strategy that produced this chunk
content_tsv	TSVECTOR	IDX	Generated column for full-text search
created_at	TIMESTAMPZ	NN	

Indexes `uq_chunk_seq` (document_version_id, sequence) UNIQUE · `idx_chunk_fts` (content_tsv) USING GIN · `idx_chunk_trgm` (content gin_trgm_ops) USING GIN · `idx_chunk_collection` (collection_id, document_id)

chunk_embeddings

Vectors, versioned by model so an embedding migration is possible.

Column	Type	Flags	Description
id	UUID	PK	
chunk_id	UUID	FK NN IDX	→ chunks.id ON DELETE CASCADE
collection_id	UUID	FK NN IDX	Denormalised — the vector query filters on it first
embedding_model_id	UUID	FK NN	→ models.id
embedding_version	SMALLINT	NN IDX	Matched against <code>collections.active_embedding_version</code> at query time
embedding	VECTOR(1536)	NN	Normalised on write; cosine distance at query time
created_at	TIMESTAMPZ	NN	

Indexes `uq_chunk_emb` (chunk_id, embedding_version) UNIQUE · `idx_emb_hnsw` (embedding vector_cosine_ops) USING hnsw WITH (m=16, ef_construction=64) · `idx_emb_filter` (collection_id, embedding_version)

ingestion_jobs

One job per document version through the pipeline.

Column	Type	Flags	Description
id	UUID	PK	
document_version_id	UUID	FK NN IDX	→ document_versions.id
collection_id	UUID	FK NN IDX	
workspace_id	UUID	FK NN IDX	

status	VARCHAR(20)	NN IDX	queued parsing chunking embedding indexed failed cancelled
stage_failed	VARCHAR(20)		Which stage threw. Diagnosis starts here
error_message	TEXT		
chunks_created	INTEGER		
embeddings_created	INTEGER		
tokens_embedded	INTEGER		Feeds the embedding cost entry
attempt_count	SMALLINT	NN	Max 3 then parked as failed. DEF 0
trace_id	VARCHAR(32)		
queued_at	TIMESTAMPTZ	NN	
started_at	TIMESTAMPTZ		
completed_at	TIMESTAMPTZ		
duration_ms	INTEGER		

Indexes `idx_ingest_status (status, queued_at)` — worker polling query

retrieval_queries

Every retrieval, with its results and scores. The raw material for retrieval evaluation.

Column	Type	Flags	Description
id	UUID	PK	
workspace_id	UUID	FK NN IDX	
user_id	UUID	FK IDX	The principal whose permissions were applied — for an agent, the delegating user
agent_run_id	UUID	FK IDX	→ agent_runs.id when retrieval was a tool call
gateway_request_id	UUID	FK IDX	→ gateway_requests.id for the generation step
query_text	TEXT	NN	
collection_ids	UUID[]	NN	Collections searched
filters	JSONB		Metadata filters applied
top_k	SMALLINT	NN	
rerank_applied	BOOLEAN	NN	DEF false
chunks_returned	SMALLINT	NN	
chunks_excluded_by_permission	SMALLINT	NN	A rising number here means permissions are misconfigured. DEF 0
top_score	NUMERIC(6,4)		Fused score of the best result. A low top score is the signal for a no-answer response
keyword_latency_ms	INTEGER		
vector_latency_ms	INTEGER		
rerank_latency_ms	INTEGER		
total_latency_ms	INTEGER		
feedback_rating	VARCHAR(20)	IDX	helpful unhelpful incorrect. NULL until a user responds
feedback_reason	TEXT		
expected_answer	TEXT		Supplied on <i>incorrect</i> feedback. Promotion source for eval datasets
promoted_to_eval_item_id	UUID	FK	→ eval_items.id once promoted
created_at	TIMESTAMPTZ	NN IDX	

Indexes `idx_rq_feedback (feedback_rating, created_at DESC) PARTIAL WHERE feedback_rating IS NOT NULL`

retrieval_citations

Which chunks were cited in which answer. Enables citation-validity scoring.

Column	Type	Flags	Description
id	UUID	PK	
retrieval_query_id	UUID	FK NN IDX	→ retrieval_queries.id ON DELETE CASCADE
chunk_id	UUID	FK NN IDX	→ chunks.id
document_id	UUID	FK NN	Denormalised — a chunk may be deleted while a citation record must survive
marker	SMALLINT	NN	The [n] the model emitted
rank_keyword	SMALLINT		Rank in the BM25 leg. NULL = not returned by keyword search
rank_vector	SMALLINT		Rank in the vector leg
fused_score	NUMERIC(6,4)	NN	RRF result
included_in_context	BOOLEAN	NN	FALSE when dropped by the context window budget
was_cited	BOOLEAN	NN	Whether the model actually referenced it. DEF false
citation_valid	BOOLEAN		NULL until scored. FALSE = the model cited a chunk that does not support the claim
flagged_bad_by_user	BOOLEAN	NN	From KB-008 feedback. Distinguishes retrieval failure from generation failure. DEF false
created_at	TIMESTAMPTZ	NN	

5. Agents & Workflows

agent_definitions

A named agent. Behaviour lives in versions.

Column	Type	Flags	Description
id	UUID	PK	
workspace_id	UUID	FK NN IDX	→ workspaces.id
slug	VARCHAR(80)	NN	
name	VARCHAR(200)	NN	
description	TEXT		
owner_user_id	UUID	FK NN	→ users.id. Accountable person, surfaced in the approval queue
production_version_id	UUID	FK	→ agent_versions.id
staging_version_id	UUID	FK	→ agent_versions.id
is_active	BOOLEAN	NN	DEF true
deleted_at	TIMESTAMPTZ		
created_at	TIMESTAMPTZ	NN	
updated_at	TIMESTAMPTZ	NN	

Indexes `uq_agent_slug (workspace_id, slug) UNIQUE`

agent_versions

An immutable agent configuration. Deployable and rollback-able.

Column	Type	Flags	Description
id	UUID	PK	
agent_definition_id	UUID	FK NN IDX	→ agent_definitions.id ON DELETE CASCADE
version_number	INTEGER	NN	
graph_definition	JSONB	NN	Nodes and edges. Node types: <code>llm</code> , <code>tool</code> , <code>condition</code> , <code>approval</code> , <code>parallel</code> , <code>end</code>
system_prompt_version_id	UUID	FK	→ prompt_versions.id
model_id	UUID	FK	→ models.id. Mutually exclusive with <code>routing_policy_id</code>
routing_policy_id	UUID	FK	→ routing_policies.id
max_steps	SMALLINT	NN	DEF 25
max_tool_calls	SMALLINT	NN	DEF 15
max_cost_usd	NUMERIC(10,4)	NN	Hard stop. An agent without a cost cap is an open invoice. DEF 1.0000
timeout_seconds	SMALLINT	NN	DEF 300
changelog	TEXT		
validation_status	VARCHAR(20)	NN	<code>valid</code> <code>invalid</code> . Set on save; invalid versions cannot be promoted
validation_errors	JSONB		Per-node errors for inline display
created_by	UUID	FK NN	→ users.id
created_at	TIMESTAMPTZ	NN	IMMUTABLE

Indexes `uq_agentver (agent_definition_id, version_number) UNIQUE`

tools

The registry of what agents can do.

Column	Type	Flags	Description
id	UUID	PK	
workspace_id	UUID	FK IDX	NULL = built-in tool available to all workspaces
slug	VARCHAR(80)	NN	knowledge_search sql_query http_request create_ticket send_notification
name	VARCHAR(200)	NN	
description	TEXT	NN	Shown to the model. Its quality directly affects tool-selection accuracy
args_schema	JSONB	NN	JSON Schema. Arguments validated before the handler runs
handler_ref	VARCHAR(200)	NN	Import path
side_effect_class	VARCHAR(20)	NN IDX	read write external destructive ✓ CHECK. Drives default approval requirements
requires_approval_default	BOOLEAN	NN	Overridable per grant. DEF true for write and destructive
runs_as_principal	BOOLEAN	NN	TRUE = executes with the delegating user's permissions. Must be TRUE for any data-access tool. DEF true
domain_allowlist	TEXT[]		For http_request. Empty array = no external access
timeout_seconds	SMALLINT	NN	DEF 30
is_active	BOOLEAN	NN	DEF true
created_at	TIMESTAMPTZ	NN	
updated_at	TIMESTAMPTZ	NN	

Indexes `uq_tool_slug (workspace_id, slug) UNIQUE NULLS NOT DISTINCT`

agent_tool_grants

Default deny. An agent with no grants has no tools.

Column	Type	Flags	Description
id	UUID	PK	
agent_version_id	UUID	FK NN IDX	→ agent_versions.id ON DELETE CASCADE
tool_id	UUID	FK NN IDX	→ tools.id
workspace_id	UUID	FK NN	Denormalised
max_calls_per_run	SMALLINT		NULL = the agent-level tool-call limit applies
requires_approval	BOOLEAN	NN	Overrides the tool default for this agent
argument_constraints	JSONB		e.g. {"schema":{"enum":["analytics_read"]}} — restricts values, not just presence
granted_by	UUID	FK NN	→ users.id. Every grant writes to audit_log
created_at	TIMESTAMPTZ	NN	

Indexes `uq_grant (agent_version_id, tool_id) UNIQUE`

agent_runs

One invocation of an agent version.

Column	Type	Flags	Description
id	UUID	PK	
workspace_id	UUID	FK NN IDX	
agent_definition_id	UUID	FK NN IDX	→ agent_definitions.id
agent_version_id	UUID	FK NN	→ agent_versions.id. Pinned, so history survives a version change

thread_id	UUID	NN IDX	LangGraph checkpoint thread. Groups resumed runs
parent_run_id	UUID	FK	→ agent_runs.id when an agent spawned a sub-agent
delegating_user_id	UUID	FK NN IDX	→ users.id. Whose permissions tools execute with — the security-critical field on this table
triggered_by	VARCHAR(20)	NN	user schedule webhook agent evaluation
trigger_ref	VARCHAR(255)		
input	JSONB	NN	
output	JSONB		
status	VARCHAR(20)	NN IDX	running awaiting_approval completed failed cancelled limit_exceeded timeout
termination_reason	VARCHAR(50)		max_steps max_cost max_tool_calls timeout rejected error
step_count	SMALLINT	NN	DEF 0
tool_call_count	SMALLINT	NN	DEF 0
input_tokens	INTEGER	NN	DEF 0
output_tokens	INTEGER	NN	DEF 0
cost_usd	NUMERIC(12,6)	NN	DEF 0. Checked against the version's cap after every step
trace_id	VARCHAR(32)	IDX	
error_message	TEXT		
started_at	TIMESTAMPTZ	NN IDX	
completed_at	TIMESTAMPTZ		
duration_ms	INTEGER		

Indexes `idx_run_agent_time (agent_definition_id, started_at DESC)` · `idx_run_status (status) PARTIAL WHERE status IN ('running', 'awaiting_approval')`

agent_steps

Every node execution. Append-only. This is the evidence that the platform is governable.

Column	Type	Flags	Description
id	UUID	PK	
agent_run_id	UUID	FK NN IDX	→ agent_runs.id ON DELETE CASCADE
workspace_id	UUID	FK NN	Denormalised
step_number	SMALLINT	NN	1-indexed
node_name	VARCHAR(100)	NN	Node in the graph definition
step_type	VARCHAR(20)	NN IDX	thought tool_call tool_result approval output error ✓ CHECK
tool_id	UUID	FK	→ tools.id
tool_arguments	JSONB		Post-validation arguments. Sensitive values redacted per GOV-001
tool_result	JSONB		Truncated above 32KB with a reference to full storage
tool_error	TEXT		
thought_text	TEXT		Model reasoning
gateway_request_id	UUID	FK	→ gateway_requests.id for the LLM call behind this step
approval_request_id	UUID	FK	→ approval_requests.id
input_tokens	INTEGER		
output_tokens	INTEGER		
cost_usd	NUMERIC(12,6)		
duration_ms	INTEGER		

created_at	TIMESTAMPTZ	NN	IMMUTABLE
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Indexes `uq_step (agent_run_id, step_number) UNIQUE`

approval_requests

A paused run waiting on a person.

Column	Type	Flags	Description
id	UUID	PK	
agent_run_id	UUID	FK NN IDX	→ agent_runs.id
workspace_id	UUID	FK NN IDX	
step_number	SMALLINT	NN	The step that triggered the pause
checkpoint_id	VARCHAR(200)	NN	LangGraph checkpoint to resume from
requested_action	VARCHAR(200)	NN	Human-readable summary, e.g. "Create ticket in OPS project"
tool_id	UUID	FK	→ tools.id
proposed_arguments	JSONB	NN	What the agent wants to do
agent_reasoning	TEXT	NN	Why. A reviewer cannot approve what they cannot evaluate
trigger_reason	VARCHAR(50)	NN	<code>approval_node</code> <code>side_effect_class</code> <code>grant_requires</code> <code>policy_requires</code>
status	VARCHAR(20)	NN IDX	<code>pending</code> <code>approved</code> <code>rejected</code> <code>modified_approved</code> <code>timed_out</code>
decided_by	UUID	FK	→ users.id
decided_at	TIMESTAMPTZ		
decision_note	TEXT		
modified_arguments	JSONB		Reviewer-edited arguments when status = <code>modified_approved</code>
expires_at	TIMESTAMPTZ	NN IDX	Auto-reject deadline. DEF now() + 24h
created_at	TIMESTAMPTZ	NN	

Indexes `idx_approval_pending (workspace_id, status, expires_at) PARTIAL WHERE status = 'pending'`

6. Evaluation

prompt_templates

A named prompt. Content lives in versions.

Column	Type	Flags	Description
id	UUID	PK	
workspace_id	UUID	FK NN IDX	
slug	VARCHAR(80)	NN	
name	VARCHAR(200)	NN	
description	TEXT		
production_version_id	UUID	FK	→ prompt_versions.id. Label pointer; rollback is a pointer move
staging_version_id	UUID	FK	→ prompt_versions.id
owner_user_id	UUID	FK NN	→ users.id
deleted_at	TIMESTAMPTZ		
created_at	TIMESTAMPTZ	NN	
updated_at	TIMESTAMPTZ	NN	

Indexes `uq_prompt_slug (workspace_id, slug) UNIQUE`

prompt_versions

Immutable prompt revisions.

Column	Type	Flags	Description
id	UUID	PK	
prompt_template_id	UUID	FK NN IDX	→ prompt_templates.id ON DELETE CASCADE
version_number	INTEGER	NN	
content	TEXT	NN	With <code>{{variable}}</code> placeholders
variables_schema	JSONB	NN	JSON Schema. Rendering fails loudly on a missing variable rather than emitting the placeholder
default_model_id	UUID	FK	→ models.id
default_temperature	NUMERIC(3,2)		
default_max_tokens	INTEGER		
changelog	TEXT		
created_by	UUID	FK NN	→ users.id
created_at	TIMESTAMPTZ	NN	IMMUTABLE

Indexes `uq_promptver (prompt_template_id, version_number) UNIQUE`

eval_datasets

A versioned test set. Immutable once referenced by a run.

Column	Type	Flags	Description
id	UUID	PK	
workspace_id	UUID	FK NN IDX	
slug	VARCHAR(80)	NN	
name	VARCHAR(200)	NN	
description	TEXT		
version	INTEGER	NN	New version on any edit after first use. Historical scores must stay comparable

target_type	VARCHAR(20)	NN	prompt rag agent routing
item_count	INTEGER	NN	DEF 0
is_locked	BOOLEAN	NN	TRUE once an eval run references it. DEF false
created_by	UUID	FK NN	→ users.id
created_at	TIMESTAMPTZ	NN	
updated_at	TIMESTAMPTZ	NN	

Indexes `uq_dataset (workspace_id, slug, version) UNIQUE`

eval_items

One test case.

Column	Type	Flags	Description
id	UUID	PK	
eval_dataset_id	UUID	FK NN IDX	→ eval_datasets.id ON DELETE CASCADE
workspace_id	UUID	FK NN	
external_ref	VARCHAR(100)		Stable ID from import, for cross-version comparison
input	JSONB	NN	Variables or question
expected_output	TEXT		
expected_behaviour	VARCHAR(50)		answer no_answer escalate refuse. Lets a no-answer be a correct result
expected_chunk_ids	UUID[]		Ground truth for retrieval metrics
expected_model_slug	VARCHAR(100)		Ground truth for routing evaluation
split	VARCHAR(10)	NN	train dev test. Reporting on the set used for iteration hides overfitting. DEF test
tags	TEXT[]		
difficulty	VARCHAR(10)		easy medium hard
source	VARCHAR(20)	NN	manual import production. Production-sourced items are the valuable ones
source_query_id	UUID	FK	→ retrieval_queries.id when promoted from feedback
created_at	TIMESTAMPTZ	NN	

Indexes `idx_item_dataset (eval_dataset_id, split)`

scorers

A scoring function with its calibration record.

Column	Type	Flags	Description
id	UUID	PK	
workspace_id	UUID	FK IDX	NULL = built-in
slug	VARCHAR(80)	NN	exact_match faithfulness precision_at_k citation_validity
name	VARCHAR(200)	NN	
scorer_type	VARCHAR(20)	NN IDX	deterministic retrieval llm_judge ✓ CHECK
handler_ref	VARCHAR(200)	NN	
config	JSONB		k for precision@k, thresholds, normalisation options
judge_model_id	UUID	FK	→ models.id. Pinned — an unpinned judge makes historical scores incomparable
judge_prompt_version_id	UUID	FK	→ prompt_versions.id. Also pinned

human_agreement_rate	NUMERIC(4,3)		Agreement with hand labels. An uncalibrated judge is a random number with good manners
calibration_sample_size	SMALLINT		
calibrated_at	TIMESTAMPZ		
passing_threshold	NUMERIC(4,3)		Score at or above which passed is TRUE
version	INTEGER	NN	Pinned on runs. DEF 1
is_active	BOOLEAN	NN	DEF true
created_at	TIMESTAMPZ	NN	

Indexes `uq_scorer (workspace_id, slug, version) UNIQUE NULLS NOT DISTINCT`

eval_runs

One dataset executed against one configuration.

Column	Type	Flags	Description
id	UUID	PK	
workspace_id	UUID	FK NN IDX	
name	VARCHAR(200)		
eval_dataset_id	UUID	FK NN IDX	→ eval_datasets.id
dataset_version	INTEGER	NN	Pinned
split	VARCHAR(10)	NN	Which split was run
target_type	VARCHAR(20)	NN	<code>prompt rag agent routing</code>
prompt_version_id	UUID	FK	→ prompt_versions.id
agent_version_id	UUID	FK	→ agent_versions.id
collection_id	UUID	FK	→ collections.id for RAG runs
model_id	UUID	FK	→ models.id
retrieval_config	JSONB		top_k, rerank, filters — pinned for reproducibility
scorer_ids	UUID[]	NN	
sample_size	SMALLINT		NULL = full split
status	VARCHAR(20)	NN IDX	<code>queued running completed failed cancelled</code>
items_total	SMALLINT	NN	
items_completed	SMALLINT	NN	DEF 0
items_failed	SMALLINT	NN	DEF 0
aggregate_scores	JSONB		Mean per scorer, pass rate, latency percentiles
total_cost_usd	NUMERIC(12,6)	NN	Visible live so an expensive sweep can be stopped. DEF 0
mlflow_run_id	VARCHAR(64)	IDX	Links to the MLflow experiment
triggered_by	UUID	FK	→ users.id
started_at	TIMESTAMPZ		
completed_at	TIMESTAMPZ		
created_at	TIMESTAMPZ	NN IDX	

Indexes `idx_evalrun_dataset (eval_dataset_id, created_at DESC)`

eval_results

Per-item, per-scorer outcome. Append-only.

Column	Type	Flags	Description
id	UUID	PK	

eval_run_id	UUID	FK NN IDX	→ eval_runs.id ON DELETE CASCADE
eval_item_id	UUID	FK NN IDX	→ eval_items.id
scorer_id	UUID	FK NN IDX	→ scorers.id
scorer_version	INTEGER	NN	Pinned
score	NUMERIC(6,4)		NULL when the scorer errored
passed	BOOLEAN		
reason	TEXT		Why this score. A bare number tells you nothing when it drops
actual_output	TEXT		
retrieved_chunk_ids	UUID[]		For retrieval metric recomputation without a re-run
gateway_request_id	UUID	FK	→ gateway_requests.id
agent_run_id	UUID	FK	→ agent_runs.id
trace_id	VARCHAR(32)		A failing item opens straight into its trace
input_tokens	INTEGER		
output_tokens	INTEGER		
cost_usd	NUMERIC(12,6)		
latency_ms	INTEGER		
error_message	TEXT		
created_at	TIMESTAMPTZ	NN	IMMUTABLE

Indexes `uq_result (eval_run_id, eval_item_id, scorer_id) UNIQUE` · `idx_result_failures (eval_run_id, passed) PARTIAL WHERE passed = false`

7. Observability

traces

Trace headers, partitioned by month. Spans hold the detail.

Column	Type	Flags	Description
trace_id	VARCHAR(32)	PK	W3C trace ID
workspace_id	UUID	FK IDX	From OTel baggage
root_span_name	VARCHAR(200)	NN	
service_name	VARCHAR(100)	NN	api worker
status	VARCHAR(20)	NN IDX	ok error
duration_ms	INTEGER	NN IDX	
span_count	SMALLINT	NN	
gateway_request_id	UUID	FK IDX	Correlation anchor across gateway, cost, agent and trace
agent_run_id	UUID	FK IDX	
started_at	TIMESTAMPTZ	NN IDX	Partition key, monthly
ended_at	TIMESTAMPTZ		

Indexes `idx_trace_ws_time` (workspace_id, started_at DESC) · `idx_trace_slow` (duration_ms DESC, started_at DESC)

spans

Individual operations. High volume, partitioned by day, 30-day retention.

Column	Type	Flags	Description
id	BIGSERIAL	PK	Insertion order matters and volume is high
trace_id	VARCHAR(32)	NN IDX	→ traces.trace_id
span_id	VARCHAR(16)	NN	
parent_span_id	VARCHAR(16)	IDX	NULL for the root
name	VARCHAR(200)	NN IDX	gateway.route, gateway.provider_call, knowledge.retrieve, agent.step
kind	VARCHAR(20)	NN	server client internal producer consumer
status	VARCHAR(20)	NN	ok error unset
status_message	TEXT		
attributes	JSONB		GenAI semantic conventions plus nexus.* namespace
gen_ai_system	VARCHAR(50)	IDX	Promoted from attributes for query performance
gen_ai_request_model	VARCHAR(100)	IDX	Promoted
gen_ai_input_tokens	INTEGER		Promoted
gen_ai_output_tokens	INTEGER		Promoted
duration_ms	INTEGER	NN IDX	
started_at	TIMESTAMPTZ	NN IDX	Partition key, daily
ended_at	TIMESTAMPTZ	NN	

Indexes `idx_span_trace` (trace_id, started_at) · `idx_span_name_time` (name, started_at DESC)

span_events

Point-in-time events on a span: retries, fallbacks, breaker transitions.

Column	Type	Flags	Description
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id	BIGSERIAL	PK	
trace_id	VARCHAR(32)	NN IDX	
span_id	VARCHAR(16)	NN IDX	
name	VARCHAR(100)	NN	retry.attempted fallback.triggered circuit.opened approval.requested
attributes	JSONB		
occurred_at	TIMESTAMPTZ	NN IDX	

metric_rollups

Pre-aggregated metrics. Every dashboard reads here, never from raw request tables.

Column	Type	Flags	Description
id	UUID	PK	
workspace_id	UUID	FK NN IDX	
grain	VARCHAR(10)	NN IDX	hour day ✓ CHECK
period_start	TIMESTAMPTZ	NN IDX	
dimension	VARCHAR(20)	NN	workspace model user api_key agent collection
dimension_id	UUID	IDX	NULL when dimension = workspace
request_count	INTEGER	NN	DEF 0
success_count	INTEGER	NN	DEF 0
error_count	INTEGER	NN	DEF 0
errors_by_class	JSONB		Counts keyed by normalised error class
input_tokens	BIGINT	NN	DEF 0
output_tokens	BIGINT	NN	DEF 0
cached_tokens	BIGINT	NN	DEF 0
cost_usd	NUMERIC(14,6)	NN	DEF 0
latency_digest	BYTEA		Serialised t-digest. Merged across buckets to compute correct percentiles — averaging stored percentiles produces a plausible wrong number
latency_p50_ms	INTEGER		Materialised for fast reads
latency_p95_ms	INTEGER		
latency_p99_ms	INTEGER		
fallback_count	INTEGER	NN	DEF 0
retrieval_hit_rate	NUMERIC(5,4)		Share of retrievals returning above the score threshold
tool_success_rate	NUMERIC(5,4)		
approval_rate	NUMERIC(5,4)		Share of approval requests approved
dimension_hash	VARCHAR(64)	NN	Idempotency key for re-runs
computed_at	TIMESTAMPTZ	NN	

Indexes `uq_rollup (workspace_id, grain, period_start, dimension, dimension_id) UNIQUE NULLS NOT DISTINCT` · `idx_rollup_query (workspace_id, grain, period_start DESC)`

alert_rules

Threshold rules and their notification routing.

Column	Type	Flags	Description
id	UUID	PK	
workspace_id	UUID	FK NN IDX	
name	VARCHAR(200)	NN	

metric	VARCHAR(50)	NN	error_rate latency_p95 cost_burn_pct circuit_open ingestion_failure_rate zero_result_rate
dimension	VARCHAR(20)		Scope the rule to a model, agent or collection
dimension_id	UUID		
comparison	VARCHAR(5)	NN	gt lt gte lte ✓ CHECK
threshold	NUMERIC(12,4)	NN	
window_minutes	SMALLINT	NN	DEF 5
consecutive_breaches	SMALLINT	NN	Suppresses single-datapoint noise. DEF 2
channel_type	VARCHAR(20)	NN	webhook email slack
channel_config	JSONB	NN	URL or address. Secrets referenced, not stored
is_active	BOOLEAN	NN	DEF true
last_fired_at	TIMESTAMPTZ		
last_resolved_at	TIMESTAMPTZ		
current_state	VARCHAR(20)	NN	ok breaching firing. DEF ok
created_at	TIMESTAMPTZ	NN	
updated_at	TIMESTAMPTZ	NN	

8. Governance & Audit

audit_log

Append-only, hash-chained, partitioned by month. The record the platform's argument rests on.

Column	Type	Flags	Description
id	BIGSERIAL	PK	Sequential — insertion order is part of the evidence
workspace_id	UUID	FK IDX	NULL for platform-level events
event_type	VARCHAR(50)	NN IDX	record.mutation auth.login auth.failed apikey.created apikey.revoked permission.granted policy.changed approval.decided pii.revealed data.exported
table_name	VARCHAR(100)	IDX	For record mutations
record_id	UUID	IDX	
operation	VARCHAR(10)		INSERT UPDATE DELETE ✓ CHECK
changed_fields	TEXT[]		UPDATE only
old_values	JSONB		NULL on INSERT
new_values	JSONB		NULL on DELETE
actor_type	VARCHAR(20)	NN	user api_key agent system
actor_id	UUID	IDX	
impersonated_by	UUID	FK	Set when an admin acted on a user's behalf
request_id	UUID	IDX	Correlates every mutation within one HTTP request
trace_id	VARCHAR(32)		
ip_address	INET		
user_agent	TEXT		
prev_hash	VARCHAR(64)		SHA-256 of the previous row in this partition
row_hash	VARCHAR(64)	NN	SHA-256 over this row's content plus prev_hash. Breaks detectably if a row is altered
created_at	TIMESTAMPTZ	NN IDX	Partition key, monthly. IMMUTABLE

Indexes `idx_audit_record (table_name, record_id, created_at DESC)` · `idx_audit_actor (actor_id, created_at DESC)` · `idx_audit_event (event_type, created_at DESC)` **RLS** `USING (false) WITH CHECK (true)` — INSERT and SELECT permitted, UPDATE and DELETE rejected by the database rather than by convention.

policies

Governance rules evaluated at gateway, tool and retrieval boundaries.

Column	Type	Flags	Description
id	UUID	PK	
workspace_id	UUID	FK NN IDX	
name	VARCHAR(200)	NN	
policy_type	VARCHAR(20)	NN IDX	model tool data content ✓ CHECK
version	INTEGER	NN	
is_current	BOOLEAN	NN	
mode	VARCHAR(10)	NN	monitor enforce ✓ CHECK. Never deploy straight to enforce on live traffic. DEF monitor
priority	SMALLINT	NN	Lower evaluates first. DEF 100
rules	JSONB	NN	Type-specific: allowed/blocked model IDs, blocked tool IDs, collection-to-provider constraints, content patterns
applies_to	JSONB		Scope: all, specific API keys, specific users,

			specific agents
deny_message	VARCHAR(300)	NN	Returned to the caller. Must name the policy — an unattributable denial is undebuggable
created_by	UUID	FK NN	→ users.id
deleted_at	TIMESTAMPTZ		
created_at	TIMESTAMPTZ	NN	

Indexes `uq_policy_ver (workspace_id, name, version) UNIQUE` · `idx_policy_eval (workspace_id, policy_type, is_current, priority) PARTIAL WHERE is_current`

policy_violations

Every denial, and every would-be denial in monitor mode. Append-only.

Column	Type	Flags	Description
id	UUID	PK	
workspace_id	UUID	FK NN IDX	
policy_id	UUID	FK NN IDX	→ policies.id
policy_version	INTEGER	NN	
enforcement_point	VARCHAR(20)	NN	gateway tool retrieval
action_taken	VARCHAR(20)	NN IDX	blocked would_block approval_required.would_block is the monitor-mode record
matched_rule	VARCHAR(200)	NN	
actor_type	VARCHAR(20)	NN	
actor_id	UUID	IDX	
gateway_request_id	UUID	FK IDX	
agent_run_id	UUID	FK IDX	
requested_resource	VARCHAR(300)		Model slug, tool slug or collection name
context	JSONB		Enough to reconstruct the decision, without the payload itself
created_at	TIMESTAMPTZ	NN IDX	IMMUTABLE

Indexes `idx_violation_policy (policy_id, created_at DESC)`

pii_detections

What was found and what was done. Never what it was. Append-only.

Column	Type	Flags	Description
id	UUID	PK	
workspace_id	UUID	FK NN IDX	
gateway_request_id	UUID	FK IDX	
gateway_message_id	UUID	FK IDX	
document_id	UUID	FK IDX	Detection during ingestion
entity_type	VARCHAR(40)	NN IDX	national_id bank_account phone email person_name address credential
char_start	INTEGER	NN	Offsets only. Storing the detected value would defeat the purpose
char_end	INTEGER	NN	
confidence	NUMERIC(4,3)	NN	
detector	VARCHAR(50)	NN	presidio regex custom_rule
redaction_rule_id	UUID	FK	→ redaction_rules.id when a custom rule matched
action_taken	VARCHAR(20)	NN	redacted_pre_provider redacted_pre_storage flagged_only

			allowlisted
vault_token	VARCHAR(64)	Reference for reversible redaction. Resolvable only with <code>governance.pii.reveal</code>	
created_at	TIMESTAMPTZ	NN IDX	IMMUTABLE

Indexes `idx_pii_type_time` (`workspace_id`, `entity_type`, `created_at` DESC)

redaction_rules

Custom patterns and false-positive allow-lists.

Column	Type	Flags	Description
id	UUID	PK	
workspace_id	UUID	FK NN IDX	
name	VARCHAR(200)	NN	
rule_type	VARCHAR(20)	NN	<code>detect</code> <code>allowlist</code> ✓ CHECK. Allowlist stops a product code being flagged as an ID number
entity_type	VARCHAR(40)	NN	
pattern	TEXT	NN	Regex, validated and length-bounded at save to prevent catastrophic backtracking
replacement	VARCHAR(100)		e.g. <code>[REDACTED_ID]</code> . DEF <code>[REDACTED]</code>
reversible	BOOLEAN	NN	Store a vault token for authorised reveal. DEF false
is_active	BOOLEAN	NN	DEF true
created_by	UUID	FK NN	→ <code>users.id</code>
created_at	TIMESTAMPTZ	NN	
updated_at	TIMESTAMPTZ	NN	

feature_flags

Runtime toggles with targeting.

Column	Type	Flags	Description
id	UUID	PK	
key	VARCHAR(100)	UQ NN	<code>HYBRID_RERANK_ENABLED</code> <code>AGENT_STUDIO_ENABLED</code> <code>PII_PRE_PROVIDER_REDACTION</code> <code>EVAL_JUDGE_SCORERS</code>
enabled	BOOLEAN	NN	Global state. DEF false
rollout_percentage	SMALLINT		0–100. NULL = all or none
workspace_ids	UUID[]		Restrict to specific workspaces. NULL = all
user_ids	UUID[]		Internal testing
description	TEXT	NN	What this controls and what to expect when it flips
updated_by	UUID	FK	→ <code>users.id</code>
created_at	TIMESTAMPTZ	NN	
updated_at	TIMESTAMPTZ	NN	

data_retention_schedules

Retention policy and legal basis, per table.

Column	Type	Flags	Description
id	UUID	PK	
table_name	VARCHAR(100)	UQ NN	

workspace_id	UUID	FK IDX	NULL = platform default; a row with a workspace overrides it
retention_days	INTEGER	NN	
strategy	VARCHAR(30)	NN	delete_row null_columns hash_identifiers aggregate_only ✓ CHECK
target_columns	TEXT[]		Columns cleared under null_columns
legal_basis	VARCHAR(100)	NN	e.g. contract_performance, legitimate_interest, regulatory_7yr
last_run_at	TIMESTAMPTZ		
next_run_at	TIMESTAMPTZ	IDX	
rows_affected_last_run	INTEGER		
approved_by	UUID	FK	→ users.id
updated_at	TIMESTAMPTZ	NN	

Alembic migration order

Later migrations reference foreign keys from earlier ones. Run in strict order.

#	Name	Tables / objects	Task
000	Extensions	vector, pgcrypto, pg_trgm	INFRA-001
001	Identity & access	users, workspaces, workspace_memberships, roles, role_assignments, api_keys, user_sessions	INFRA-002
002	Gateway & routing	providers, models, model_deployments, routing_policies, routing_rules, gateway_requests, gateway_messages, fallback_events	INFRA-002
003	Cost & quotas	pricing_rates, cost_entries, budgets, quota_windows, rate_limit_buckets	INFRA-002
004	Knowledge Hub	collections, collection_grants, documents, document_versions, chunks, chunk_embeddings, ingestion_jobs, retrieval_queries, retrieval_citations	INFRA-003
005	Agents	agent_definitions, agent_versions, tools, agent_tool_grants, agent_runs, agent_steps, approval_requests	INFRA-003
006	Evaluation	prompt_templates, prompt_versions, eval_datasets, eval_items, scorers, eval_runs, eval_results	INFRA-003
007	Observability	traces, spans, span_events, metric_rollups, alert_rules	INFRA-003
008	Governance	audit_log, policies, policy_violations, pii_detections, redaction_rules, feature_flags, data_retention_schedules	INFRA-003
009	Triggers	set_updated_at() on all tables with the column; log_audit() on users, api_keys, routing_policies, policies, agent_tool_grants, collection_grants, role_assignments	INFRA-003
010	Immutability & partitioning	RLS on the seven append-only tables; range partitions on audit_log and traces (monthly), spans and gateway_requests (daily); partition-creation function	INFRA-003

Key relationships at a glance

From	Card.	To	Business rule
workspaces	1:N	everything	The tenancy boundary. Every business row carries workspace_id.
users	N:M	workspaces	Via workspace_memberships. A user may belong to several.
routing_policies	1:N	routing_rules	Ordered, first match wins.

gateway_requests	1:N	gateway_messages	One row per message, subject to logging mode.
gateway_requests	1:N	fallback_events	One row per model switch. Usually zero.
gateway_requests	1:1	cost_entries	Exactly one cost entry per completed request.
models	1:N	pricing_rates	Effective-dated, non-overlapping.
collections	1:N	documents	A document belongs to exactly one collection.
documents	1:N	document_versions	Re-upload versions rather than overwrites.
document_versions	1:N	chunks	Re-chunking replaces the set transactionally.
chunks	1:N	chunk_embeddings	One per embedding version, enabling zero-downtime model migration.
collections	1:N	collection_grants	Retrieval joins here. Default deny.
retrieval_queries	1:N	retrieval_citations	Every retrieved chunk, cited or not.
agent_definitions	1:N	agent_versions	Labels point at versions; rollback is a pointer move.
agent_versions	1:N	agent_tool_grants	Default deny. No grant, no tool.
agent_runs	1:N	agent_steps	Full execution trace, append-only.
agent_runs	1:N	approval_requests	One per pause.
agent_runs	N:1	users (delegating)	Tools execute as this principal, never as a service account.
prompt_templates	1:N	prompt_versions	Immutable revisions.
eval_datasets	1:N	eval_items	Locked once referenced by a run.
eval_runs	1:N	eval_results	One row per item per scorer.
traces	1:N	spans	Spans partitioned by day, 30-day retention; rollups kept indefinitely.
gateway_requests	1:1	traces	Joined on <code>trace_id</code> . One request ID resolves request, cost, trace and agent run.